

T10 — CheckMyMetal-style IZI ion-identity verdict

Local Zheng-criteria heuristic · Mg²⁺ excluded for both ions · K⁺ has best raw bond-length fit but low CN + biology

Ca²⁺ A 98 · mean bond 2.887 Å · CN 3 · B 42.73

Element	IZI (bond)	Verdict
Mg ²⁺	16.35	EXCLUDED — bonds 16σ too long for Mg ²⁺ canonical (2.07 Å)
Ca ²⁺	3.25	consistent with partial-occupancy framing
Na ⁺	2.44	possible
K ⁺	0.44	best raw match · CN low · protein is known Ca ²⁺ -enzyme

Na⁺ B 98 · mean bond 2.773 Å · CN 3 · B 22.85

Element	IZI (bond)	Verdict
Mg ²⁺	14.06	EXCLUDED
Na ⁺	1.87	borderline
K ⁺	0.13	better bond-length fit than Na ⁺ (open question)

Net verdict: agent’s modelling of Ca²⁺ and Na⁺ is defensible

Mg²⁺ alternative excluded for both ions. Ca²⁺ for chain A is the best biological + geometric fit (Asp33 is the canonical SNase Ca²⁺-binder).

Open follow-up: K⁺ has the best raw bond-length fit. Need crystallisation buffer information to close.